

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 3 – Mock Interview

|                  |                                                                                                                                  |               |                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------|
| Course Code:     | DSA0305                                                                                                                          | Course Title: | Natural Language Processing        |
| CO Assessed:     | CO5 – Articulation                                                                                                               | Assessment:   | Assessment Tool 3 – Mock Interview |
| Weightage:       | 30% of CIA                                                                                                                       |               |                                    |
| CO5 Statement:   | Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5) |               |                                    |
| Student Name:    | N.Adi Narayana                                                                                                                   | Reg. No.:     | 192525346                          |
| Section / Batch: | AIML                                                                                                                             | Date:         | 19-08-2026                         |

### Code of Conduct

I N.Adi Narayana (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_Adi\_\_\_\_\_

### Instructions

- Start the mock interview by entering the provided ChatGPT prompt exactly as given in the Annexure A in the next page. Enter your Name and Register Number when requested.
- Answer all 10 interview questions independently and in your own words using mobile chatgpt voice. Do not copy content from external sources or use AI-generated responses during the interview.
- Provide detailed and meaningful answers with appropriate technical terminology, examples, and justifications wherever applicable. One-line answers may lead to lower scores.
- Complete the interview in a single session. Ensure that all 10 questions are answered before requesting the final evaluation and assessment report.
- Submit the final ChatGPT-generated report printout, including the interview questions, your responses, question-wise rubric evaluation, and the Student Average Final Mark, as proof of completion.
- Duration: 30 minutes.

| Section / Topic                                      | Focus                                                                                                                 | Question No. |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------|
| Discourse & Reference Resolution                     | Understanding discourse, anaphora resolution, reference tracking, and contextual interpretation in text               | Q1           |
| Text Coherence & Discourse Structure                 | Analyzing coherence relations, discourse organization, and maintaining meaningful text flow                           | Q2           |
| Dialog Acts & Interpretation                         | Identifying dialog acts, speaker intentions, and conversational meaning                                               | Q3           |
| Dialog and Conversational Agents                     | Architecture, components, and working of conversational agents/chatbots                                               | Q4           |
| Language Generation Architecture                     | Stages of Natural Language Generation (NLG), pipeline architecture, and applications                                  | Q5           |
| Surface Realization & Discourse Planning             | Transforming semantic representations into grammatically correct sentences and planning discourse                     | Q6           |
| Machine Translation – Transfer Approach              | Translation process using transfer-based methods and practical applications                                           | Q7           |
| Machine Translation – Interlingua Approach           | Language-independent representation and multilingual translation framework                                            | Q8           |
| Statistical Approaches in NLP                        | Statistical language models, probability-based language generation, and machine translation                           | Q9           |
| Integrated Case Study (Analytical & Problem Solving) | Real-world scenario involving discourse analysis, conversational agents, language generation, and machine translation | Q10          |

## Annexure A – Mock Interview

---

CHATGPT PROMPT: copy and paste in your chatgpt

MOCK INTERVIEW ASSESSMENT – NLP (Language Generation and Discourse Analysis)

Act as an expert Technical Interview Panel for Computer Science and Engineering students.

Interview Domain:

Language Generation and Discourse Analysis

Topics:

- Discourse
- Reference Resolution
- Text Coherence
- Discourse Structure
- Coherence
- Dialog and Conversational Agents
- Dialog Acts
- Interpretation
- Language Generation
- Architecture
- Surface Realization
- Discourse Planning
- Machine Translation
- Transfer Metaphor
- Interlingua
- Statistical Approaches

Interview Instructions:

Step 1:

Ask the student to enter:

Student Name:

Register Number:

Step 2:

Conduct a technical interview consisting of exactly 10 questions.

Rules:

- Ask only one question at a time.
- Wait for the student's response.
- Ask conceptual, analytical, application-based, and problem-solving questions.
- Include at least:
  - 2 questions on Discourse and Coherence
  - 2 questions on Conversational Agents and Dialog Acts
  - 2 questions on Language Generation
  - 2 questions on Machine Translation
  - 2 analytical or scenario-based questions

Step 3:

Store every question and answer.

Step 4:

After Question 10, evaluate the student using the following rubric.

## RUBRIC

### Technical Knowledge (25)

Level 4:

Demonstrates strong and accurate subject knowledge with in-depth understanding

Level 3:

Shows good knowledge with minor gaps

Level 2:

Basic knowledge with noticeable errors

Level 1:

Lacks fundamental technical knowledge

### Problem Solving (25)

Level 4:

Solves problems logically and applies concepts effectively in real scenarios

Level 3:

Applies concepts with moderate accuracy

Level 2:

Limited problem-solving ability; requires guidance

Level 1:

Unable to solve or apply concepts

### Analytical Thinking (20)

Level 4:

Analyzes questions critically and provides well-structured logical answers

Level 3:

Provides reasonable analysis with some structure

Level 2:

Superficial or partially structured responses

Level 1:

No analytical thinking demonstrated

### Communication (15)

Level 4:

Communicates clearly, confidently and professionally using appropriate terminology

Level 3:

Communicates well with minor hesitation

Level 2:  
Communication lacks clarity or confidence

Level 1:  
Unable to communicate effectively

Confidence (15)

Level 4:  
Displays high confidence, positive attitude and professional behavior throughout

Level 3:  
Shows good confidence with minor issues

Level 2:  
Low confidence and inconsistent behavior

Level 1:  
Lacks confidence and professionalism

## MARK CALCULATION

Level 4 = 100% of weightage

Level 3 = 75% of weightage

Level 2 = 50% of weightage

Level 1 = 25% of weightage

Technical Knowledge = 25 marks

Problem Solving = 25 marks

Analytical Thinking = 20 marks

Communication = 15 marks

Confidence = 15 marks

Calculate:

Final Score = Sum of all rubric scores

Step 5:

Generate the final report in the exact format below.

---

## MOCK INTERVIEW ASSESSMENT REPORT

Student Name:

Register Number:

## INTERVIEW QUESTIONS AND RESPONSES

Q1:

Question Asked:

Student Response:

Q2:  
Question Asked:  
Student Response:

Q3:  
Question Asked:  
Student Response:

Q4:  
Question Asked:  
Student Response:

Q5:  
Question Asked:  
Student Response:

Q6:  
Question Asked:  
Student Response:

Q7:  
Question Asked:  
Student Response:

Q8:  
Question Asked:  
Student Response:

Q9:  
Question Asked:  
Student Response:

Q10:  
Question Asked:  
Student Response:

#### RUBRIC EVALUATION

| Criterion           | Weightage Assigned | Level   | Marks Awarded |
|---------------------|--------------------|---------|---------------|
| Technical Knowledge | 25                 | Level X | XX            |
| Problem Solving     | 25                 | Level X | XX            |
| Analytical Thinking | 20                 | Level X | XX            |
| Communication       | 15                 | Level X | XX            |
| Confidence          | 15                 | Level X | XX            |

#### SUMMARY TABLE

| Technical Knowledge (25) | Problem Solving (25) | Analytical Thinking (20) | Communication (15) | Confidence (15) | SubQ Total | Student Avg Final Mark |
|--------------------------|----------------------|--------------------------|--------------------|-----------------|------------|------------------------|
| Level X                  | Level X              | Level X                  | Level X            | Level X         | XX.X       | XX.X                   |

#### OVERALL FEEDBACK

Strengths:

- List strengths

Areas for Improvement:

- List improvements

Interview Readiness:

- Excellent / Good / Average / Needs Improvement

End the report after the Interview Readiness section.

After evaluating each answer, do NOT display the marks immediately.

Store the evaluation internally.

After Question 10 is completed:

1. Generate a Question-wise Evaluation Table for all 10 questions.

2. For each question assign:

- Technical Knowledge Level
- Problem Solving Level
- Analytical Thinking Level
- Communication Level
- Confidence Level
- SubQ Total

3. Calculate SubQ Total using:

Level 4 = 100% weightage

Level 3 = 75% weightage

Level 2 = 50% weightage

Level 1 = 25% weightage

4. Calculate Student Avg Final Mark as:

Average of all 10 SubQ Totals

5. Display:

- Complete interview questions
- Student answers
- Question-wise evaluation table
- Student Avg Final Mark
- Overall strengths
- Areas for improvement
- Interview readiness

# ANNEXURE B – MOCK INTERVIEW ANSWER SHEET

---

The following section presents complete, exam-ready answers for all 10 interview questions listed in Annexure A, covering Discourse Analysis, Conversational Agents, Natural Language Generation and Machine Translation.

## **Q1. What is discourse? Explain reference resolution and anaphora resolution with an example.**

Discourse is a sequence of connected sentences or utterances that together form a meaningful communication. In Natural Language Processing, discourse analysis goes beyond individual sentences and studies how information, entities, events, and meanings are connected across sentences.

For example:

*"Ravi went to the library. He borrowed a book."*

If we analyze only the second sentence, we know that "He" is a male pronoun, but we do not know who it refers to. By using the previous sentence as discourse context, we can determine:

Ravi → He

This process is called reference resolution.

## **Anaphora Resolution**

Anaphora resolution is the task of finding the antecedent of an expression that refers back to something previously mentioned.

Example:

*"Priya bought a laptop. She uses it every day."*

There are two references:

She → Priya  
it → laptop

The system uses several constraints:

### **1. Gender Agreement**

The pronoun and antecedent should agree in gender where applicable.

He → Ravi  
She → Priya



## 2. Number Agreement

Singular and plural forms should be compatible.

```
he → singular  
they → plural
```

## 3. Recency

Recently mentioned entities are often preferred as possible antecedents.

## 4. Grammatical Role

Entities appearing as subjects are often more salient.

## 5. Semantic Compatibility

The antecedent must make sense in the context. For example:

*"The student bought a laptop. He switched it on."*

Here:

```
He → student  
it → laptop
```

A laptop can be switched on, while the student is not the intended object of "it."

## 6. Discourse Coherence

The selected references should produce a consistent interpretation throughout the discourse.

Therefore, reference resolution is important for question answering, chatbots, machine translation, summarization, information extraction, and conversational AI.

## Simple Processing Flow

```
Text → Identify referring expressions → Find candidate entities → Apply  
grammatical constraints → Apply semantic compatibility → Check discourse context  
→ Select antecedent → Resolved discourse
```

Thus, discourse provides the context required for NLP systems to understand references that cannot be interpreted from an individual sentence alone.

## Q2. What is text coherence? Explain discourse structure and coherence relations.

Text coherence refers to the logical and meaningful connection between different sentences or parts of a document. A coherent text does not simply contain grammatically correct sentences; its sentences are connected in a way that produces an understandable overall meaning.

Consider:

*"It rained heavily. Therefore, the roads were flooded. Schools were closed."*

The sentences have a clear relationship:

Heavy Rainfall → Road Flooding → School Closure

This makes the paragraph coherent.

## Important Coherence Relations

### 1. Cause–Effect

One event causes another. Example:

*"It rained heavily, so the roads were flooded."*

Rainfall → Flooding

### 2. Sequence

Events occur in a particular order. Example:

*"The student completed the assignment. Then she submitted it."*

Complete Assignment → Submit Assignment

### 3. Elaboration

A later sentence provides additional information. Example:

*"The company launched an AI chatbot. The chatbot can answer customer questions automatically."*

The second sentence elaborates on the first.

### 4. Contrast

Two pieces of information are presented as contrasting. Example:

*"The model is expensive, but it provides high accuracy."*

The word "but" indicates contrast.

## Discourse Structure

A discourse can be represented as a tree, where a Cause-Effect relation joins sentence S1 to a Sequence relation joining S2 and S3. For:

S1 = Heavy rainfall caused flooding.  
S2 = Schools were closed.  
S3 = Students attended classes online.

The structure is:

Rainfall → Flooding → School Closure → Online Classes

Coherence is particularly important in NLP applications such as automatic summarization, document understanding, question answering, text generation, machine translation, and conversational systems.

Without coherence, a generated response may contain grammatically correct sentences but still fail to communicate a meaningful overall message.

### **Q3. What is a dialogue act? Explain different dialogue acts with an example.**

A dialogue act represents the communicative purpose of a particular utterance. It tells the conversational system what the speaker is trying to do through the utterance.

For example:

*"Can you book a train ticket for me?"*

The user is not simply providing information. The user is making a Request.

Common dialogue acts include:

| Dialogue Act | Meaning                             | Example                    |
|--------------|-------------------------------------|----------------------------|
| Request      | Ask the system to perform an action | "Book a ticket."           |
| Question     | Ask for information                 | "What time is the train?"  |
| Inform       | Provide information                 | "I want to go to Chennai." |
| Confirmation | Confirm information / action        | "Your ticket is booked."   |
| Action       | Perform an operation                | Book the ticket            |

Consider this conversation:

User: "Can you book a train ticket for me?" → Dialogue Act = REQUEST  
Agent: "Where would you like to travel?" → Dialogue Act = QUESTION  
User: "I want to go to Chennai." → Dialogue Act = INFORM

```
System books the ticket → Dialogue Act = ACTION  
System: "Your ticket has been booked." → Dialogue Act = CONFIRMATION
```

Therefore, the dialogue proceeds as:

```
REQUEST → QUESTION → INFORM → ACTION → CONFIRMATION
```

Dialogue-act recognition helps a conversational agent understand the user's intention rather than merely identifying individual words.

#### **Q4. Explain the architecture and working of a conversational agent.**

A conversational agent is an NLP-based system that interacts with users through natural language. It can be implemented as a chatbot, customer-support assistant, booking assistant, voice assistant, or AI agent.

### **General Architecture**

```
User → Input Processing → Natural Language Understanding → Intent + Entity  
Extraction → Dialogue Manager → Context / State Tracking → Knowledge Base /  
Tools → Response Generation → NLG → Surface Realization → Final Response
```

#### **1. Input Processing**

The system receives text or speech. Example:

*"Book a train to Chennai."*

#### **2. Natural Language Understanding**

The system extracts:

```
Intent = BookTrain  
Destination = Chennai
```

#### **3. Dialogue Manager**

The dialogue manager maintains the current conversation state. For example:

```
Destination = Chennai  
Date = Missing
```

The system then knows that it needs to ask for the date.

#### **4. Knowledge Base or External Tools**

The system may access databases, APIs, search systems, booking systems, and CRM systems.

#### **5. Response Generation**

The system decides what should be communicated.

## 6. Natural Language Generation

The semantic response is converted into natural language.

## 7. Surface Realization

The final grammatical sentence is produced. For example:

*"What date would you like to travel?"*

## Importance

A conversational agent must maintain context. If the user says "Book a ticket to Chennai." and later says "Tomorrow.", the system must understand that "tomorrow" refers to the travel date rather than treating it as an independent statement.

Therefore, conversational agents combine language understanding, dialogue management, context tracking, external tools, and language generation.

## Q5. Explain Natural Language Generation and its architecture.

Natural Language Generation (NLG) is the process of converting structured information, semantic representations, or system decisions into natural-language text.

For example, a system may internally represent:

```
Agent = Student  
Action = Buy  
Object = Book  
Tense = Past
```

NLG converts this into:

*"The student bought a book."*

## NLG Architecture

```
Semantic Representation → Content Planning → Discourse Planning → Sentence  
Planning → Lexical Selection → Surface Realization → Final Text
```

### 1. Content Planning

Determines what information should be included. For example, if a delivery system knows:

```
Status = Delayed  
Arrival = Tomorrow
```

the NLG system decides that both pieces of information should be communicated.

## 2. Discourse Planning

Determines the organization and relationship between information. For example:

```
Cause → Effect  
Problem → Solution  
Event → Explanation
```

## 3. Sentence Planning

Determines how information should be grouped into sentences.

## 4. Lexical Selection

Selects suitable words. For example:

```
BUY + Past → bought
```

## 5. Surface Realization

Converts the planned structure into grammatically correct language.

NLG is used in chatbots, virtual assistants, automatic report generation, weather reports, recommendation systems, summarization, business intelligence systems, and automated customer support.

The main objective is to produce text that is meaningful, coherent, grammatically correct and appropriate to the context.

### **Q6. What is surface realization? Explain it with an example.**

Surface realization is the stage of Natural Language Generation where a semantic or syntactic representation is converted into a grammatically correct natural-language sentence.

Suppose the input is:

```
Action = Buy  
Agent = Student  
Object = Book  
Tense = Past
```

First, lexical items are selected:

```
Student → student  
Buy → buy  
Book → book
```

The tense is then applied:

buy → bought

The sentence structure is Subject + Verb + Object, and the system adds determiners:

The student + bought + a book

Final output:

*"The student bought a book."*

## Surface Realization Pipeline

Semantic Representation → Lexical Selection → Morphological Processing → Word Ordering → Determiner Selection → Agreement Checking → Punctuation → Final Sentence

The system must verify subject–verb agreement, tense, word order, morphology, articles/determiners, and punctuation. For example:

Incorrect: "The student buy a book yesterday."  
Correct: "The student bought a book yesterday."

The realization stage therefore ensures that the final generated sentence is grammatically acceptable.

## Q7. Explain the Transfer Approach in Machine Translation.

The Transfer Approach is a Machine Translation method in which the source language is first analyzed and then transformed into a target-language-oriented representation before generating the target sentence.

### Architecture

Source Language → Source Analysis → Intermediate Representation → Language-Pair Transfer → Target Representation → Target Generation → Target Language

For example:

*"The boy is playing football."*

The source analysis identifies:

Subject = boy  
Verb = playing  
Object/Activity = football  
Tense = present  
Aspect = continuous

The transfer component then converts this structure according to the grammatical requirements of the target language. Finally, the target-language generator produces the translated sentence.

### Advantages

Handles grammatical differences explicitly. Allows language-specific rules. Provides greater linguistic control. Useful for domains with well-understood grammar.

**Limitations**

Requires language-pair-specific transfer rules. Developing rules can be time-consuming. Maintaining many language pairs becomes difficult. For example, English → Tamil, English → Hindi, and English → French may require different transfer rules.

Therefore, transfer-based MT provides explicit linguistic control but has scalability limitations when many languages are involved.

**Q8. Explain the Interlingua approach and compare it with the Transfer approach.**

The Interlingua approach represents the meaning of a source sentence in a language-independent intermediate representation.

**Architecture**

Source Language → Semantic Analysis → Interlingua → Target Language Generation → Target Language

For example:

*"The boy is playing football."*

can be represented as:

EVENT = PLAY  
AGENT = BOY  
ACTIVITY = FOOTBALL  
TENSE = PRESENT  
ASPECT = CONTINUOUS

This representation is independent of English. The same semantic representation could be used to generate English, Tamil, Hindi, and French.

**Transfer vs Interlingua**

| Feature                     | Transfer               | Interlingua          |
|-----------------------------|------------------------|----------------------|
| Intermediate representation | Language-pair specific | Language-independent |
| Transfer rules              | Required               | Minimized/avoided    |
| Multilingual scalability    | Lower                  | Higher               |



|                          |                |                            |
|--------------------------|----------------|----------------------------|
| Semantic representation  | Moderate       | Strong                     |
| Language-pair dependency | High           | Lower                      |
| Main challenge           | Transfer rules | Designing rich Interlingua |

Transfer: English → English-specific representation → Tamil transfer → Tamil  
Interlingua: English → Interlingua → Tamil, and English → Interlingua → Hindi

The main advantage of Interlingua is that the same meaning representation can support multiple languages. Its major challenge is designing an expressive representation capable of capturing the semantic and linguistic requirements of different languages.

## **Q9. What are statistical approaches in NLP? Explain statistical language models and their role in Machine Translation.**

Statistical approaches in NLP use probability and data-driven methods to make predictions about language. A statistical language model estimates the probability of a sequence of words. For example:

$P(\text{"The student bought a book"})$

A model can compare different possible sequences and assign probabilities based on patterns learned from data.

### **Statistical Language Model**

A simplified model can estimate  $P(w_1, w_2, \dots, w_n)$ . For example, it may determine that "The student bought a book" is more probable than an unnatural word sequence.

### **Statistical Machine Translation**

In statistical Machine Translation, the system considers possible target-language translations and assigns probabilities. A simplified objective is:

Best Translation =  $\text{argmax}_{\text{Target Source}} P(\text{Target Source})$

The system can combine translation probability, language-model probability, fluency, and lexical compatibility. For example:

| Candidate     | Score |
|---------------|-------|
| Translation A | 0.92  |
| Translation B | 0.81  |

|               |      |
|---------------|------|
| Translation C | 0.65 |
|---------------|------|

The highest-scoring candidate is selected.

## Advantages

Learns patterns from data. Can handle linguistic variation. Useful for ranking alternatives. Does not require every linguistic rule to be manually written.

## Limitations

Depends heavily on training data. Rare expressions may be difficult. Errors can occur when the training data does not represent the target domain. Pure statistical methods may not explicitly represent discourse meaning.

A hybrid architecture can combine symbolic semantic representation with statistical scoring:

```
Semantic Representation → Candidate Generation → Statistical Scoring → Best Candidate → Final Output
```

Thus, statistical approaches are useful for predicting and ranking natural-language outputs.

## **Q10. Analytical Case Study: Design an NLP system for a multilingual customer-support chatbot.**

*"My order was delayed because of the rain. Can you tell me when it will arrive?"*

I would design an integrated NLP pipeline combining discourse analysis, reference resolution, dialogue-act recognition, conversational-agent architecture, NLG and machine translation.

## Overall Architecture

```
User Input → Language Detection → Text Preprocessing → Discourse Analysis → Reference Resolution → Dialogue Act Detection → Intent Recognition → Entity Extraction → Dialogue Management → Order Database / API → Semantic Response → Discourse Planning → Natural Language Generation → Translation if Needed → Surface Realization → Final Response
```

## Step 1 – Discourse Analysis

The first sentence, "My order was delayed because of the rain," expresses a Cause–Effect relationship:

```
Rain → Order Delay
```

The second sentence asks about the expected arrival, so the entire discourse is connected:

```
Rain → Delay → Question about Arrival
```

## Step 2 – Reference Resolution

The second sentence contains "when it will arrive." Possible antecedents include order and rain. The system applies semantic compatibility:

```
Order + arrive → valid  
Rain + arrive → invalid
```

Therefore, it → order. This is an example of reference resolution using semantic context.

## Step 3 – Dialogue Act Recognition

The sentence "Can you tell me when it will arrive?" is a Question, so Dialogue Act = QUESTION. The underlying intent is:

```
Intent = Track Order / Get Delivery Time
```

## Step 4 – Entity Extraction

The system identifies Entity = Order, Cause = Rain, Event = Delay, and Requested Information = Arrival Time. The structured representation may be:

```
ORDER_STATUS( order = current_order, status = delayed, requested_information =  
arrival_time )
```

## Step 5 – Dialogue Management

The dialogue manager checks whether enough information is available. If the system already knows the user's order (Order ID = Available), it can directly query the order-management system. If the order ID is unavailable, the chatbot may ask, "Could you please provide your order number?" This shows why context and state tracking are important.

## Step 6 – External Information

The chatbot queries its order database/API. Suppose the result is:

```
Status = Delayed  
Expected Arrival = Tomorrow
```

## Step 7 – Semantic Response

The response can be represented as:

```
DELIVERY_STATUS( Order = CurrentOrder, Status = Delayed, EstimatedArrival =  
Tomorrow )
```

## Step 8 – Discourse Planning

The system organizes the response logically: first the current situation, then the expected arrival.

Delay → Expected Arrival

## Step 9 – Natural Language Generation

The NLG system generates:

*"Your order was delayed because of the rain, but it is expected to arrive tomorrow."*

This response maintains the discourse context and answers the user's question.

## Step 10 – Translation

If the user requests another language, the semantic representation can be passed to an Interlingua-based translation system:

Semantic Representation → Interlingua → Target Language → Surface Realization

This separates the underlying meaning from the surface language.

## Step 11 – Final Validation

| Requirement                  | Result |
|------------------------------|--------|
| Reference resolved           | ✓      |
| Correct dialogue act         | ✓      |
| Intent identified            | ✓      |
| Cause–effect maintained      | ✓      |
| Order information preserved  | ✓      |
| Arrival information included | ✓      |
| Response coherent            | ✓      |
| Grammar correct              | ✓      |
| Translation possible         | ✓      |

Therefore, the integrated NLP system can understand the user's message, maintain discourse context, identify the user's intention, retrieve the required information and generate an appropriate response.

## **FINAL REVISION SHEET**

For the interview, remember these 10 core answers:

| Q  | Topic                            | One-line Core Idea                                  |
|----|----------------------------------|-----------------------------------------------------|
| 1  | Discourse & Reference Resolution | Use context and constraints to identify antecedents |
| 2  | Text Coherence                   | Identify logical relations between sentences        |
| 3  | Dialogue Acts                    | Identify the communicative purpose of an utterance  |
| 4  | Conversational Agents            | NLU + dialogue manager + tools + NLG                |
| 5  | NLG Architecture                 | Meaning → planning → realization → text             |
| 6  | Surface Realization              | Semantic representation → grammatical sentence      |
| 7  | Transfer MT                      | Source analysis → transfer → target generation      |
| 8  | Interlingua MT                   | Language-independent meaning representation         |
| 9  | Statistical NLP                  | Use probability to predict/rank language            |
| 10 | Case Study                       | Integrate all the above components                  |

## Rubrics for Calculation

| Criteria            | Weightage | Level 4 (Exemplary)                                                               | Level 3 (Proficient)                             | Level 2 (Developing)                               | Level 1 (Beginning)                   |
|---------------------|-----------|-----------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------|---------------------------------------|
| Technical Knowledge | 25        | Demonstrates strong and accurate subject knowledge with in-depth understanding    | Shows good knowledge with minor gaps             | Basic knowledge with noticeable errors             | Lacks fundamental technical knowledge |
| Problem Solving     | 25        | Solves problems logically and applies concepts effectively in real scenarios      | Applies concepts with moderate accuracy          | Limited problem-solving ability; requires guidance | Unable to solve or apply concepts     |
| Analytical Thinking | 20        | Analyzes questions critically and provides well-structured, logical answers       | Provides reasonable analysis with some structure | Superficial or partially structured responses      | No analytical thinking demonstrated   |
| Communication       | 15        | Communicates clearly, confidently, and professionally with proper terminology     | Communicates well with minor hesitation          | Communication lacks clarity or confidence          | Unable to communicate effectively     |
| Confidence          | 15        | Displays high confidence, positive attitude, and professional behavior throughout | Shows good confidence with minor issues          | Low confidence; inconsistent behavior              | Lacks confidence and professionalism  |

| Q. No. / Criteria Level | Problem Understanding | Algorithm Design | Use of Control Structures | Pseudocode Structure | Completeness | Sub. Total | Total % |
|-------------------------|-----------------------|------------------|---------------------------|----------------------|--------------|------------|---------|
| 1                       |                       |                  |                           |                      |              |            |         |
| 2                       |                       |                  |                           |                      |              |            |         |
| 3                       |                       |                  |                           |                      |              |            |         |
| 4                       |                       |                  |                           |                      |              |            |         |
| 5                       |                       |                  |                           |                      |              |            |         |
| 6                       |                       |                  |                           |                      |              |            |         |
| 7                       |                       |                  |                           |                      |              |            |         |
| 8                       |                       |                  |                           |                      |              |            |         |
| 9                       |                       |                  |                           |                      |              |            |         |
| 10                      |                       |                  |                           |                      |              |            |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |